

LIUZHOU, China

WMO: 590460

Lat: 24.3628N

Lon: 109.4600E

Elev: 307

StdP: 97.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	4.6	-5.7	2.4	10.3	-3.2	3.0	9.4	6.4	6.4	5.5	7.9	2.5	0	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	35.2	25.8	34.4	25.7	33.6	25.6	27.2	32.3	26.8	31.8	26.5	31.3	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.1	29.5	25.6	21.6	29.3	25.3	21.2	29.1	88.5	32.8	86.7	31.9	85.2	31.4	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	4.9	4.2	DB	2.2	37.1	1.7	0.9	0.9	37.7	-0.1	38.3	-1.1	38.8	-2.3	39.4
			WB	0.4	27.8	1.3	0.7	-0.5	28.3	-1.2	28.7	-1.9	29.1	-2.9	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	10.9	13.1	16.4	21.7	25.7	28.0	29.5	29.4	27.7	23.8	18.7	13.3
	DBStd	7.39	4.30	5.16	4.31	4.03	2.78	2.17	1.92	1.97	2.71	3.33	3.75	3.54
	HDD10.0	75	40	22	3	0	0	0	0	0	0	0	1	10
	HDD18.3	703	232	161	90	16	0	0	0	0	0	4	39	161
	CDD10.0	4298	69	109	201	352	487	539	604	602	529	428	263	112
	CDD18.3	1885	3	16	30	118	229	289	346	344	280	174	52	4
	CDH23.3	19483	6	67	126	784	2058	3173	4341	4267	3078	1314	256	13
CDH26.7	7986	0	11	17	190	712	1253	2044	2012	1310	398	37	1	
Wind	WSAvg	1.8	1.8	1.8	1.7	1.9	2.0	2.0	2.1	1.8	1.8	1.8	1.6	1.7
Precipitation	PrecAvg	1516	63	47	102	132	247	328	191	164	77	59	60	47
	PrecMax	2299	187	132	320	305	386	630	415	522	257	197	262	163
	PrecMin	963	5	3	6	34	121	118	54	32	4	2	3	5
	PrecStd	309	41	31	59	64	69	148	106	102	57	51	58	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	28.3	28.8	32.1	33.9	35.0	36.5	36.2	35.5	33.1	29.5	25.0
		MCWB	18.2	20.6	21.9	23.8	25.2	25.9	26.1	25.5	25.0	23.3	20.9	19.4
	2%	DB	21.7	25.3	26.3	30.4	32.6	33.8	35.3	35.2	34.2	31.6	27.7	22.6
		MCWB	16.6	19.2	20.8	23.3	24.7	25.9	26.1	25.7	25.0	22.7	20.1	16.6
	5%	DB	19.6	23.1	24.5	28.8	31.5	32.9	34.3	34.3	33.2	30.3	26.3	20.6
		MCWB	15.1	18.5	19.9	22.8	24.4	25.7	26.0	25.7	24.8	22.2	19.4	14.7
	10%	DB	17.4	21.1	22.9	27.4	30.4	31.8	33.3	33.4	32.2	29.0	24.6	18.6
		MCWB	13.4	17.2	19.0	22.3	24.2	25.6	25.9	25.6	24.4	21.4	18.6	13.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	21.2	22.8	25.2	27.0	27.5	27.7	27.6	26.7	25.3	23.3	21.4
		MCDB	22.8	26.8	27.1	29.8	31.8	32.5	33.4	32.6	31.9	30.3	26.4	24.1
	2%	WB	17.8	20.0	21.9	24.4	26.2	27.0	27.2	27.1	26.3	24.5	22.1	18.3
		MCDB	20.8	24.3	25.4	28.7	30.7	31.7	32.6	32.1	31.2	29.3	26.3	21.6
	5%	WB	15.9	18.9	20.9	23.7	25.6	26.6	26.7	26.7	26.0	23.8	21.2	15.8
		MCDB	18.9	22.6	23.8	27.6	29.8	31.0	31.9	31.6	30.6	28.3	25.2	18.8
	10%	WB	13.9	17.6	19.6	23.1	25.1	26.2	26.4	26.4	25.5	23.0	19.8	14.1
		MCDB	16.8	20.9	22.2	26.7	28.9	30.3	31.3	31.1	30.2	27.3	23.3	17.6
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.4	5.3	5.8	6.3	5.7	6.3	6.5	6.8	6.8	6.8	6.3
		MCDBR	9.7	9.1	8.5	8.1	8.2	7.2	7.7	7.7	7.9	8.3	9.0	9.8
		MCWBR	5.0	4.4	4.1	3.2	2.7	2.0	1.7	1.8	1.9	2.7	3.5	4.7
	5% WB	MCDBR	7.9	8.1	7.0	6.9	6.5	6.0	6.6	6.8	7.0	7.2	6.9	7.6
		MCWBR	4.9	4.6	4.3	3.6	3.0	2.3	2.1	2.2	2.1	3.0	3.6	4.8
Clear-Sky Solar Irradiance	taub	0.623	0.652	0.775	0.741	0.599	0.567	0.534	0.604	0.638	0.695	0.597	0.616	
	taud	1.622	1.613	1.430	1.534	1.868	1.970	2.059	1.850	1.743	1.579	1.730	1.661	
	Ebn at Noon	626	650	598	636	732	751	775	723	684	609	644	609	
	Edh at Noon	233	250	314	288	206	185	169	208	226	255	207	216	
All-Sky Solar Radiation	RadAvg	1.91	2.16	2.46	3.18	3.88	3.95	4.85	4.77	4.42	3.73	2.85	2.30	
	RadStd	0.48	0.46	0.38	0.41	0.40	0.54	0.44	0.32	0.34	0.36	0.49	0.47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	-1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	N/A	+45

Nomenclature: See separate page